

1. **Limit at 0** (2 points)

$$\text{Let } f(x) = \begin{cases} x^2 + 4, & x < 0 \\ \sqrt{x}, & x \geq 0 \end{cases}.$$

2. **Find b for continuity at 5** (2 points)

$$\text{Let } f(x) = \begin{cases} 2x - 3, & x < 5 \\ \frac{2}{3}x + b, & x \geq 5 \end{cases}. \text{ Determine } b \text{ so } \lim_{x \rightarrow 5} f(x) \text{ exists.}$$

3. **Limit at 2** (2 points)

$$\text{Let } f(x) = \begin{cases} \frac{3}{2}x + 7, & x < 2 \\ 3x + 4, & x \geq 2 \end{cases}. \text{ Evaluate } \lim_{x \rightarrow 2} f(x).$$

4. **Limit at 4** (2 points)

$$\text{Let } f(x) = \begin{cases} \sin\left(\frac{\pi}{8}x\right), & x < 4 \\ \frac{1}{4}x, & x \geq 4 \end{cases}. \text{ Evaluate } \lim_{x \rightarrow 4} f(x).$$

5. **Limit at 1 (infinite one-sided)** (2 points)

$$\text{Let } f(x) = \begin{cases} (x - 2)^2, & x < 1 \\ \frac{1}{x - 1}, & x \geq 1 \end{cases}. \text{ Evaluate } \lim_{x \rightarrow 1} f(x).$$

6. **Limit at -5** (2 points)

$$\text{Let } f(x) = \begin{cases} \frac{3}{5}x + 2, & x < -5 \\ 2x + \frac{8}{3}, & x \geq -5 \end{cases}. \text{ Evaluate } \lim_{x \rightarrow -5} f(x).$$

7. **Find b for limit at -6 to exist** (2 points)

Let $f(x) = \begin{cases} x^2 + b, & x < -6 \\ \frac{1}{2}x + 9, & x \geq -6 \end{cases}$. Determine b so $\lim_{x \rightarrow -6} f(x)$ exists.

8. **Find m for limit at 3 to exist** (2 points)

Let $f(x) = \begin{cases} 2x + 7, & x < 3 \\ mx - 4, & x \geq 3 \end{cases}$. Determine m so $\lim_{x \rightarrow 3} f(x)$ exists.

9. **Find m, b so the limit at 4 equals 10** (3 points)

Let $f(x) = \begin{cases} mx + 2b, & x < 4 \\ 5x + b, & x \geq 4 \end{cases}$. Determine m and b so $\lim_{x \rightarrow 4} f(x) = 10$.

10. **Two-sided limit** (2 points)

Evaluate $\lim_{x \rightarrow 0} \lfloor x \rfloor$.

11. **Asymptotic ratio** (3 points)

Evaluate $\lim_{x \rightarrow 0^+} \frac{\lfloor 1/x \rfloor}{1/x}$.

12. **Scaled floor** (3 points)

Compute $\lim_{x \rightarrow 0^+} x \lfloor 1/x \rfloor$.

13. **Oscillatory difference** (4 points)

Discuss existence of $\lim_{x \rightarrow 0^+} (\lfloor 1/x \rfloor - \frac{1}{x})$ and find its $\limsup$ and $\liminf$.

14. **Symmetric floor quotient** (4 points)

Evaluate $\lim_{x \rightarrow 0} \frac{\lfloor x \rfloor + \lfloor -x \rfloor}{x}$, if it exists.

15. **Limit at an integer point** (3 points)

Let $n \in \mathbb{Z}$. Evaluate $\lim_{x \rightarrow n} \lfloor x \rfloor$.

16. **Parameter question** (5 points)

Determine all real numbers a for which the two-sided limit $\lim_{x \rightarrow 0} \frac{\lfloor ax \rfloor}{x}$ exists (as a finite real number). Justify.

17. **Limit involving the fractional part function** (4 points)

Let $\{t\} = t - \lfloor t \rfloor$ denote the fractional part. Evaluate $\lim_{x \rightarrow 0^+} x \{1/x\}$.

18. **Gaussian decay: limits of $x^n e^{-x^2}$** (3 points)

For any fixed integer $n \geq 0$, evaluate $\lim_{x \rightarrow \infty} x^n e^{-x^2}$.

19. **Gaussian integral** (4 points)

Prove that for $a > 0$, $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$.

20. **Fourier transform of a Gaussian** (5 points)

Using the convention $\mathcal{F}\{f\}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$, compute $\mathcal{F}\{e^{-ax^2}\}(k)$ for $a > 0$.

21. **Convolution of two Gaussians** (4 points)

Let $g_{\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/(2\sigma^2)}$. Show that $g_{\sigma_1} * g_{\sigma_2} = g_{\sqrt{\sigma_1^2 + \sigma_2^2}}$.

22. **Narrowing Gaussian and pointwise vs. integral limits** (5 points)

Let $G_{\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/(2\sigma^2)}$ for $\sigma > 0$. (a) For fixed $x \neq 0$, compute $\lim_{\sigma \rightarrow 0^+} G_{\sigma}(x)$. (b) Compute $\lim_{\sigma \rightarrow 0^+} \int_{-\varepsilon}^{\varepsilon} G_{\sigma}(x) dx$ for any fixed $\varepsilon > 0$.

23. **Uniform convergence failure for scaled Gaussians** (3 points)

Consider $f_n(x) = n e^{-n^2 x^2}$ on $\mathbb{R}$. Show that $f_n \rightarrow 0$ pointwise but the convergence is not uniform on $\mathbb{R}$.

